18.IMPLEMENT PERCEPTRON BASED IRIS CLASSIFICATION

1. Dataset 1 – Iris Setosa vs Versicolor

SepalLength	SepalWidth	PetalLength	PetalWidth	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.0	2.2	4.0	1.0	Versicolor
5.9	3.0	4.2	1.5	Versicolor
5.5	3.5	1.3	0.2	Setosa
6.1	2.8	4.0	1.3	Versicolor

Program

```
import pandas as pd
from sklearn.linear_model import Perceptron
df = pd.read_csv("iris1.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = Perceptron(max_iter=1000, random_state=1)
model.fit(X, y)
print("Predicted:", model.predict(X))
print("Accuracy:", model.score(X, y))
```

Output

Predicted: ['Setosa' 'Setosa' 'Versicolor' 'Versicolor' 'Setosa' 'Versicolor']

Accuracy: 1.0

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2. Dataset 2 – Iris Setosa vs Virginica

SepalLength	SepalWidth	PetalLength	PetalWidth	Species
5.0	3.4	1.5	0.2	Setosa
4.8	3.1	1.6	0.2	Setosa
6.3	2.9	5.6	1.8	Virginica
6.5	3.0	5.8	2.2	Virginica
5.2	3.5	1.5	0.2	Setosa
6.7	3.1	5.6	2.4	Virginica

Program

```
import pandas as pd
from sklearn.linear_model import Perceptrondf = pd.read_csv("iris2.csv")
X = df[["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]]
y = df["Species"]
p = Perceptron(max_iter=1000, random_state=2)
p.fit(X, y)
print("Predictions:")
print(p.predict(X))
print("Accuracy:", p.score(X, y))
```

Output

Predictions:

['Setosa' 'Setosa' 'Virginica' 'Virginica' 'Setosa' 'Virginica']

Accuracy: 1.0

3. Dataset 3 – Setosa and Versicolor

Sepal	Width	Petal	PWidth	Class
5.1	3.5	1.4	0.2	Setosa
5.0	3.4	1.5	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa
5.8	2.7	4.1	1.0	Versicolor
6.0	2.9	4.5	1.5	Versicolor
5.6	2.9	3.6	1.3	Versicolor

Program

```
import pandas as pd
from sklearn.linear_model import Perceptron
data = pd.read_csv("iris3.csv")
X = data.iloc[:, 0:4]
y = data["Class"]
model = Perceptron(max_iter=500, eta0=1)
model.fit(X, y)
print("Predicted Classes:")
print(model.predict(X))
print("Accuracy =", model.score(X, y))
```

Output

```
Predicted Classes:
['Setosa' 'Setosa' 'Setosa' 'Versicolor' 'Versicolor' 'Versicolor']

Accuracy = 1.0
```

4. Dataset 4 – Versicolor vs Virginica

SepalLength	SepalWidth	PetalLength	PetalWidth	Species
5.8	2.7	4.1	1.0	Versicolor
6.0	2.9	4.5	1.5	Versicolor
5.7	2.8	4.1	1.3	Versicolor
6.3	3.3	6.0	2.5	Virginica
6.5	3.0	5.8	2.2	Virginica
6.7	3.1	5.6	2.4	Virginica

Program

```
import pandas as pd

from sklearn.linear_model import Perceptron

df = pd.read_csv("iris4.csv")

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

model = Perceptron(max_iter=1000, random_state=4)

model.fit(X, y)

pred = model.predict(X)

print("Actual:", list(y))

print("Predicted:", list(pred))

print("Accuracy:", model.score(X, y))
```

Output

```
Actual: ['Versicolor', 'Versicolor', 'Versicolor',
    'Virginica', 'Virginica', 'Virginica']


Predicted: ['Versicolor', 'Versicolor', 'Versicolor',
```

'Virginica', 'Virginica', 'Virginica']

Accuracy: 1.0

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5. Dataset 5 – Three Iris Classes

SepalLength	SepalWidth	PetalLength	PetalWidth	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
5.9	3.0	4.2	1.5	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.3	3.3	6.0	2.5	Virginica
6.5	3.0	5.8	2.2	Virginica

Program

```
import pandas as pd
from sklearn.linear_model import Perceptron
df = pd.read_csv("iris5.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = Perceptron(max_iter=1000, random_state=5)
model.fit(X, y)
print("Predicted Classes:")
print(model.predict(X))
Print("Accuracy:", model.score(X, y))
```

Output

Predicted Classes:

['Setosa' 'Setosa' 'Versicolor' 'Versicolor'

'Virginica' 'Virginica']

Accuracy: 1.0